CANALRUNG · RISK REGISTER AND HAZARD ANALYSIS

CanalRung risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CNR-RSK-001	Version	0.1
Status	Draft	Date	2026-11-05
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/09-risk.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First risk register, preliminary hazard analysis and design FMEA (kit 1.9.1 upgrade), for option A; hazards of options B and C noted where they differ

CanalRung risk register and hazard analysis

The top risks are a rescue that fails when it is needed (a rope, rung or anchor that is weaker than the paper numbers, with the thinnest margins in the rope and the saturated-soil footing), a bystander who enters the water, and a cost that stays over the USD 500 target; none has been tested. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (not caught).

> **Safety:** All scores are the author's first-pass judgement from paper design, not test or incident data. CanalRung is a research and educational design, never certified life-saving equipment, and nothing here shows it is safe. Never enter the water to rescue someone.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Concept choice not made	Amish has not decided among options A, B and C (CNR-DDR-004)	Work stays on A; B and C modelled but unused	3	2	Amish	Decision requested in CNR-PLN-001
P2	Cost stays over the target	USD 605.20 against USD 500; footing is USD 143	Fewer stations per district	4	3	Design lead	Savings listed in CNR-DEC-001; cost is reported, `budget_usd` is not raised
P3	Unverified prices	No supplier prices could be checked on 2026-11-05	Cost off by an unknown amount	4	3	Design lead	Mark "to assess"; get quotes at TRL 4
P4	Rope rating not proven	9 mm type B strength taken from catalogue class values; margin 4.08 against 4	Rope or knot fails in a proof load	3	5	Design lead	Proof load with CalRig at TRL 4; require a declared minimum of 18 kN
P5	No co-design partner yet	Kittitas Reclamation District and the Water Safety Council of Fresno County named, not contacted	Needs, heights and soils stay assumed	4	4	Amish	Contact when allowed; plan in CNR-PLN-001
P6	Regulatory route unknown	Classification of the station is to assess (CNR-REG-001 G3)	A product claim could be unlawful	3	5	Regulatory reviewer	Keep the "research and educational prototype" wording
P7	Test facility and lifeguards unavailable	Tank trials need supervised water	TRL 4 delayed	3	3	Project lead	Plan only; no testing at TRL 3
P8	Price or part drift	Prices and part availability change	The BOM goes stale	3	2	Design lead	Re-check cited prices before any release

Preliminary hazard analysis

Table 2. Hazards (option A; B and C noted).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	A bystander enters the water to help	Urge to rescue; the ladder is slow to deploy	Drowning of the rescuer	5	3	Front decal "do not enter the water"; pictogram on the lid; throw and call	High until trials show the throw works
H2	The person is entangled in the ropes in a current	Ropes and knots in moving water	Drowning	5	2	Blunt-tip knife in the lid; no loose throw line (CNR-DDR-001 D7); ropes end at a weight, not a loop	Medium
H3	A rung, rope or anchor fails under the climber	Weaker than calculated; ageing; a bad knot	Person falls back, injury	5	2	Rung at 0.56 of yield, rope 4.08, anchor factors 1.53 to 3.58 (CNR-CAL-001); proof loads at TRL 4	Medium until proof loaded
H4	The thrown weight hits the person	Poor throw	Blunt injury	3	2	Soft shot pouch of 0.5 kg; no hard parts thrown (PR-010)	Low
H5	The ladder lies on the wall but the person cannot climb it	Cold, tired, a child; slick wall	Person stays in the water	5	3	335 mm pitch, 70 mm hand room at stand-offs, floating rungs; trials	High until trials; B or C might serve such a person better
H6	Rope burn or cut to the thrower or to the person pulling	Rope runs over the hand	Hand injury	2	3	Edge protectors; gloves not required; user guidance	Low
H7	A child hangs on or plays with the ladder	Always-ready equipment	Fall or entanglement	3	2	Ladder folded in a latched box with a tamper tag; no lock so it opens in an emergency	Medium
H8	A sharp or hard edge hurts someone	Cut tube ends, box rim	Cuts	2	2	Rung ends capped, edges broken in the model, bushed holes	Low
H9	A bystander slips on the wet bank while throwing	Wet, steep bank edge	Fall into the canal	4	3	Post set 1,000 mm back from the coping; throw from behind the post; guidance	Medium
H10	Option B: a rigid ladder falls onto a person in the water	Lowering over the edge	Blunt injury	4	3	None designed; handling and lanyard only	High (option B)

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H11	Option C: crank handle spins back on the cranker	Pawl fails or is released under load	Hand or face injury	3	3	Ratchet and pawl in the model; no brake designed	High (option C)

Design FMEA (first pass, P3)

Table 3. Failure modes for option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Rung	Carry the climber's weight	Rung bends or buckles	The ladder gives way under one foot	Over-load; cold-worked tube	4	2	3	24	Proof-load a rung (TRL 4)
F2	Side rope at a knot	Carry the load into the anchor	Rope breaks at the knot	The ladder falls	Knot weakens the rope; wet; abrasion	5	2	4	40	Proof-load a roped rung; inspect each season
F3	Knot and seizing	Hold the rung in place	Knot slips or seizing loosens	A rung drops down the rope	Wet, UV, wrong knot	4	3	3	36	Show the knot in the build plan; inspect
F4	Footing	Hold the anchor	Footing pulls out	Whole station fails	Soft or saturated soil (factor 1.59); crest of a slope cuts the passive wedge (STEM S4, S5)	5	3	4	60	Site pull test; deeper footing proposed (CNR-DEC-001, open decision 3)
F5	Eye nut and bolt	Link ladder to post	Nut or bolt fails	The ladder is lost	Wrong grade; side loading	5	1	3	15	Buy marked parts; load in line
F6	Float sleeve	Float the rung	Foam soaks or degrades	Rung sinks	UV; cut; open-cell foam	3	3	3	27	Closed-cell grade only; inspect
F7	Folded ladder in the box	Pay out cleanly	Tangle	Throw fails	Fold order; wind	4	3	4	48	Fold test and pictogram at TRL 4
F8	Aluminium mounting plate	Clamp the box on the post	Galvanic corrosion at the plate	Box loosens	Zinc and aluminium couple, wet (STEM S11)	2	3	2	12	Watch at TRL 4
F9	Lid latch	Open without tools	Latch jams or tag is broken	Delay in an emergency	Ice, dirt, vandalism	5	2	3	30	No lock; tamper tag; inspection
F10	Stand-off block	Hold the rung off the wall	Block cracks or bolt loosens	Hand room lost	UV, load	3	3	3	27	HDPE; inspect
F11	Edge protector	Keep the rope off the coping	Slips off	Rope abraded to 50 percent strength (CNR-CAL-001 E2)	Movement	4	3	3	36	Check fit and retention at TRL 4
F12	Rope bush		Bush cracks or crushes		UV, load; wet nylon yields near	4	3	3	36	Proof load at TRL 4; inspect

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
		Protect the rope in the rung hole		Rope chafes; knot pulls through	35 MPa against 32.8 MPa from a foot at the rung end (STEM S9)					
F13	Throw weight	Carry the ladder end down	Snags or splits	Ladder hangs short	Rough edge; shot spills	3	2	2	12	Check the pouch seams at TRL 4
F14	Eye nut	Link ladder to post	Eye nut loaded off its axis fails or bends	The ladder is lost	Rope pulls 14 to 34 degrees off the thread; the rating is in line only (STEM S23)	5	3	4	60	Get the off-axis rating or use a swivel hoist ring (CNR-DEC-001, open decision 5)

RPN above 40 or severity 5 with detection 4 or more are the items to act on first: F4 (60), F14 (60), F7 (48) and F2 (40). Each action is paper work for TRL 4 and is not started here.

Sources

- CNR-CAL-001 for the strength, anchor and edge figures; CNR-STM-001 for S3, S5, S10, S11 and S12; CNR-DDR-001 for D7 (no throw line).
- Severity and likelihood scales are the author's judgement; no incident database was used.